Status	Finished
Started	Wednesday, 21 October 2026, 7:27 PM
Completed	Wednesday, 21 October 2026, 7:35 PM
Duration	8 mins
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By comparing each element in the list sequentially ✓
- b. By dividing the list into halves
- c. By sorting the list first
- d. By using a hash function

Question **2**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(1) ✓
- b. O(n log n)
- c. O🥐
- d. O(log n)

Question **3**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. log n
- b. n/2
- c. n ✓
- d. 1

Question **4**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Hash search

b. Both (1) & (3)

c. Binary search

d. Linear search ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements

b. When the search algorithm finishes

c. When the element is found in the list ✓

d. When the list is sorted

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted

b. The search stops

c. The search continues from the beginning

d. The search continues in the left or right sublist ✓

Question **7**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the first element in the list

b. Sort the list

c. Divide the list into two equal parts

d. Compare the target element with the middle element in the list ✓

Question **8**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is somewhere in the middle of the array ✓

b. When the item is not the array at all

c. When the item is the last element in the array

d. Item is the last element in the array or item is not there at all

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)

b. O🥊

c. O(1)

d. O(log n) ✓

Question **10**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500

b. 1000

c. 100

d. 10 ✓

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.

b. Linear search is more efficient than binary search on large lists.

c. Linear search can be applied to both sorted and unsorted lists. ✓

d. Linear search divides the list into halves.

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. In the entire list
c. At the beginning of the list
d. In the left sublist ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. The first element is returned
c. An error is raised
d. The search is considered unsuccessful ✓

Question **14**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted
b. The search ends successfully ✓
c. The search continues in the right sublist
d. The search continues in the left sublist

Question **15**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Interpolation search
b. Hash search
c. Binary search
d. Linear search ✓